CSE 311:
Data Communication

Instructor:

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Analysis of Signals and Systems

Properties of Fourier Transform

Time scaling or dilation property $$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

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$$= \frac{1}{a} \int_{-\infty}^{\infty} g(\tau) \exp\left[-j2\pi\left(\frac{f}{a}\right)\tau\right] d\tau$$ assuming a > 0, and $at = \tau$

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$$= \frac{1}{a} G\left(\frac{f}{a}\right)$$

$$F[g(at)] = \frac{-1}{a} G(\frac{f}{a})$$ assuming a < 0

Therefore, $g(at) \Leftrightarrow \frac{1}{|a|} G(\frac{f}{a})$

Properties of Fourier Transform

Significance of Time scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|} G(\frac{f}{a})$$

- If $a > 1$, $g(at)$ means compression in time which increases the signal frequency
- Compression in time results in spectral expansion by same factor

Properties of Fourier Transform

Significance of Time scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|} G(\frac{f}{a})$$

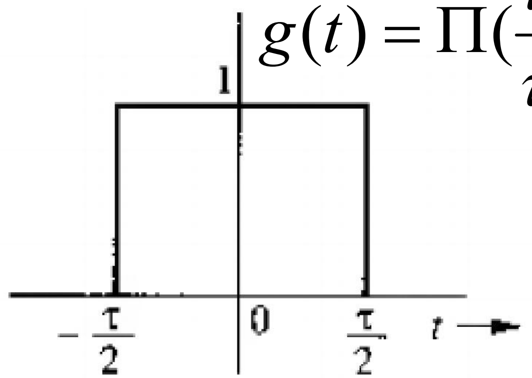
- If $a > 1$, $g(at)$ means compression in time which increases the signal frequency
- Compression in time results in spectral expansion by same factor
- Expansion in time ($a < 1$) results in spectral compression
- If $g(t)$ is wider, spectrum is narrower
- Bandwidth (spectrum) is inversely proportional to signal duration (width)

Properties of Fourier Transform

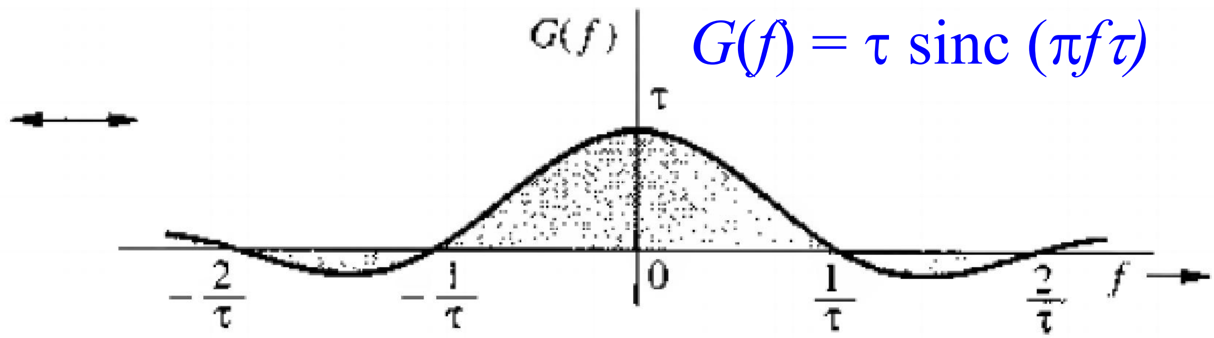
Significance of Time scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

$$g(t) = \Pi\left(\frac{t}{\tau}\right)$$



$G(f) = \tau \operatorname{sinc}(\pi f \tau)$



Properties of Fourier Transform

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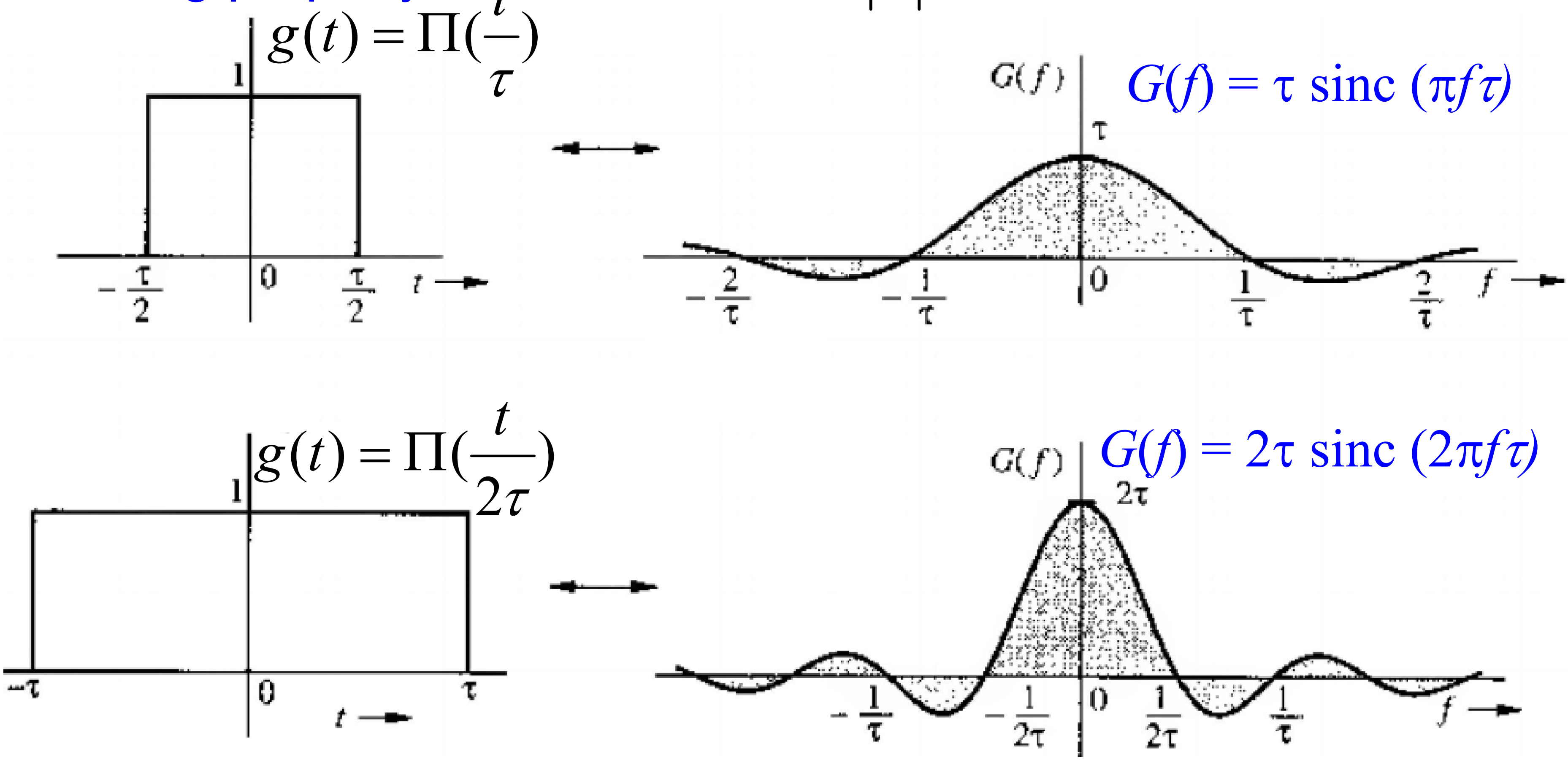

Properties of Fourier Transform

Reflection property $g(-t) \Leftrightarrow G(-f)$

Directly results assuming a = -1 in scaling property

$$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{f}{a}\right)$$

Properties of Fourier Transform

Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$

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Properties of Fourier Transform

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$$= \int_{-\infty}^{\infty} g(\tau)e^{-j2\pi f(t_0+\tau)}d\tau$$ assuming $t = t_0 + \tau$

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$$= e^{-j2\pi f t_0}\int_{-\infty}^{\infty} g(\tau)e^{-j2\pi f \tau}\,d\tau$$

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$$= e^{-j2\pi ft_0}\int_{-\infty}^{\infty} g(\tau)e^{-j2\pi f\tau}d\tau$$

$$= e^{-j2\pi ft_0}G(f)$$

Properties of Fourier Transform

Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$

- Delaying the signal by t_0 seconds DOES NOT change the amplitude spectrum, but changes phase spectrum by $-2\pi f t_0$

Properties of Fourier Transform

Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$

- Delaying the signal by t_0 seconds DOES NOT change the amplitude spectrum, but changes phase spectrum by $-2\pi f t_0$
- If $g(t)$ is synthesized by Fourier components (sinusoids of certain amplitudes and phases), $g(t-t_0)$ can be synthesized by same sinusoids delayed by t_0.

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df$$

$$= \int_{-\infty}^{\infty} [G(f)\cos 2\pi f t + jG(f)\sin 2\pi f t] df$$

Properties of Fourier Transform

Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$

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- If $g(t)$ is synthesized by Fourier components (sinusoids of certain amplitudes and phases), $g(t-t_0)$ can be synthesized by same sinusoids delayed by t_0.

Assume the cosine signal cos $2\pi f t$ delayed by t_0

$$\cos 2\pi f(t-t_0) = \cos(2\pi f t - 2\pi f t_0)$$

Properties of Fourier Transform

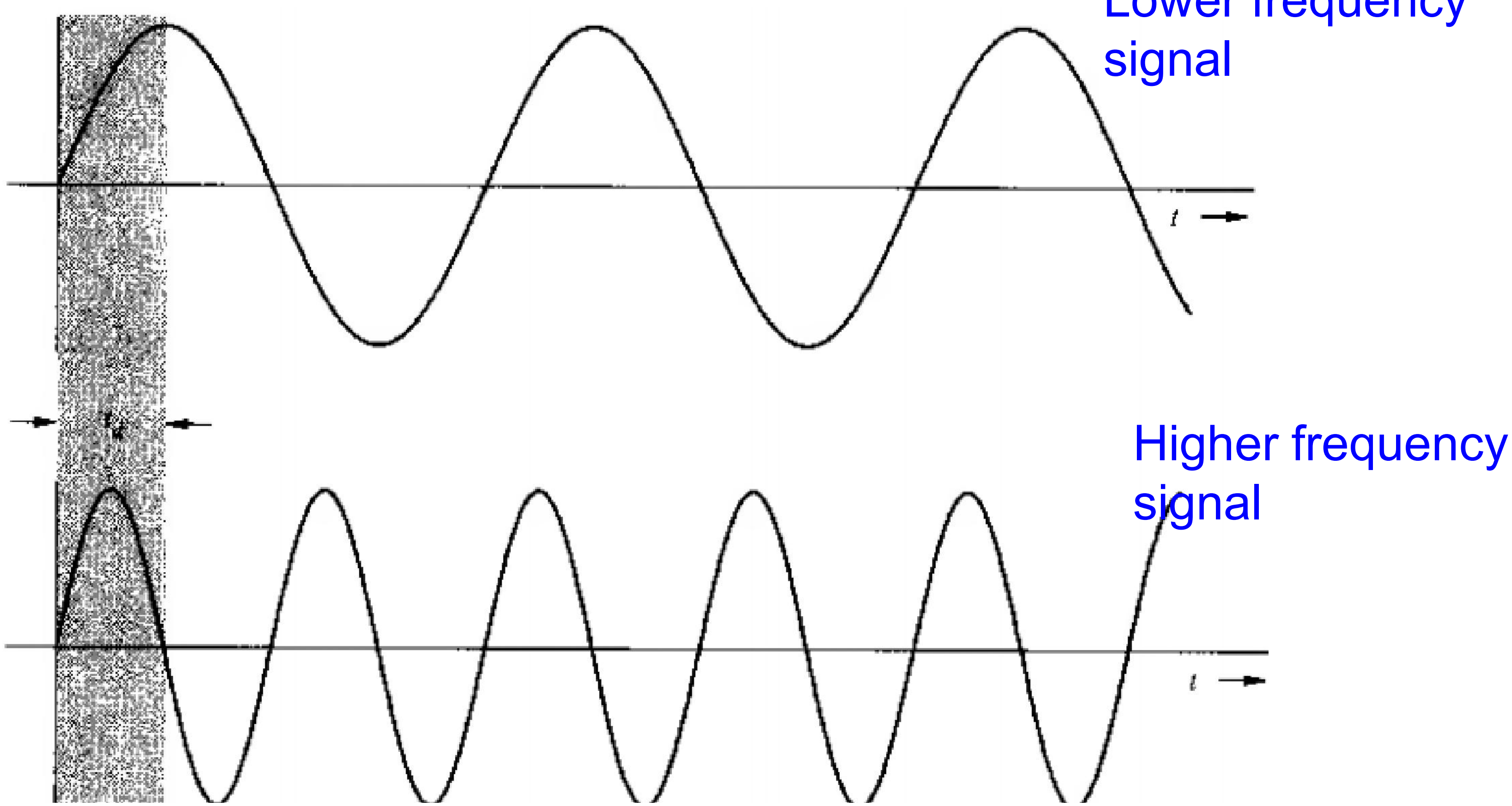
Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$

$$\cos 2\pi f(t-t_0) = \cos(2\pi f t - 2\pi f t_0)$$

- Phase changes is linear function of f
- Higher frequency components undergoes proportionately higher phase shift

Properties of Fourier Transform

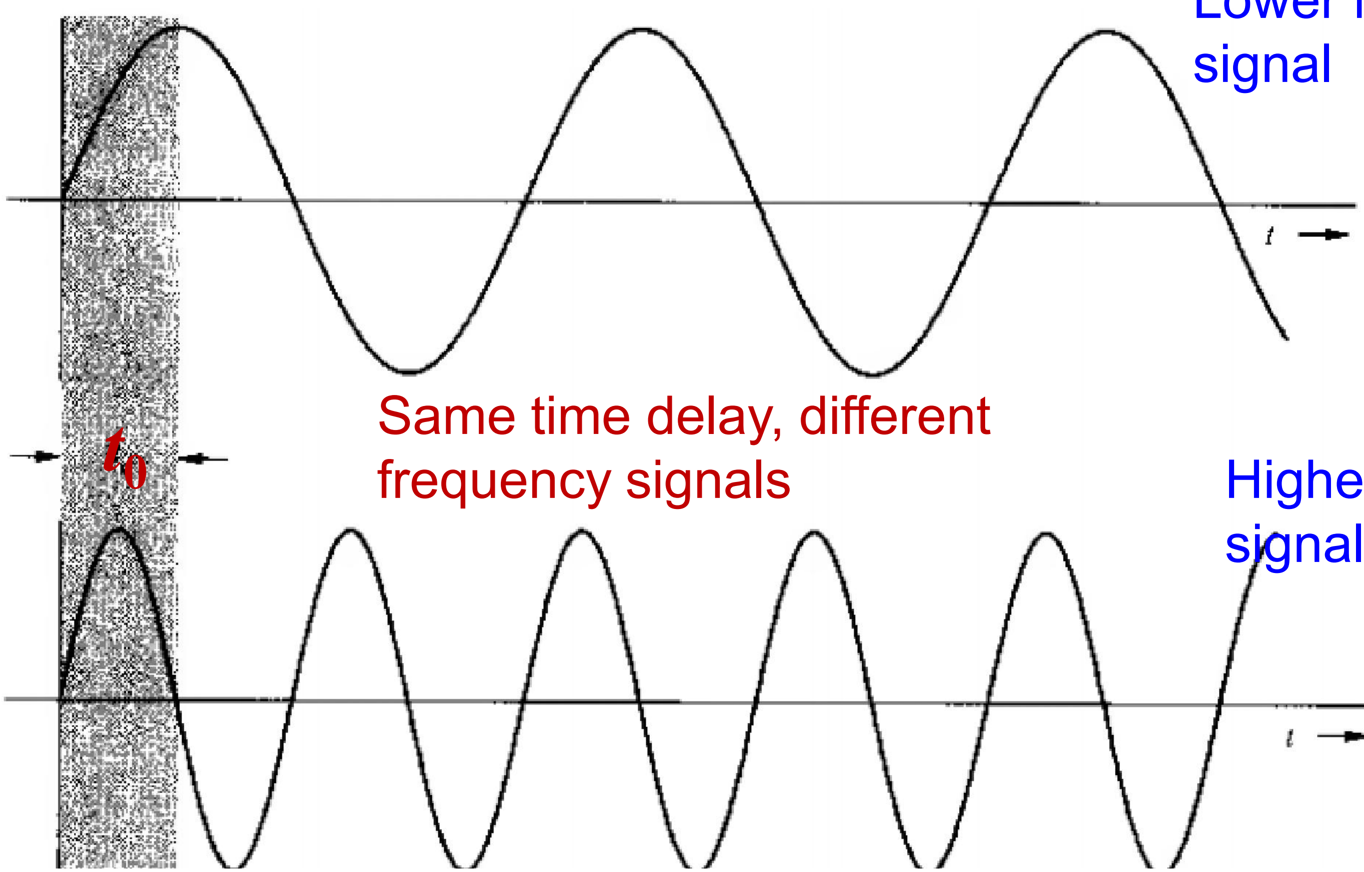
Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$



Properties of Fourier Transform

Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$

Lower frequency signal

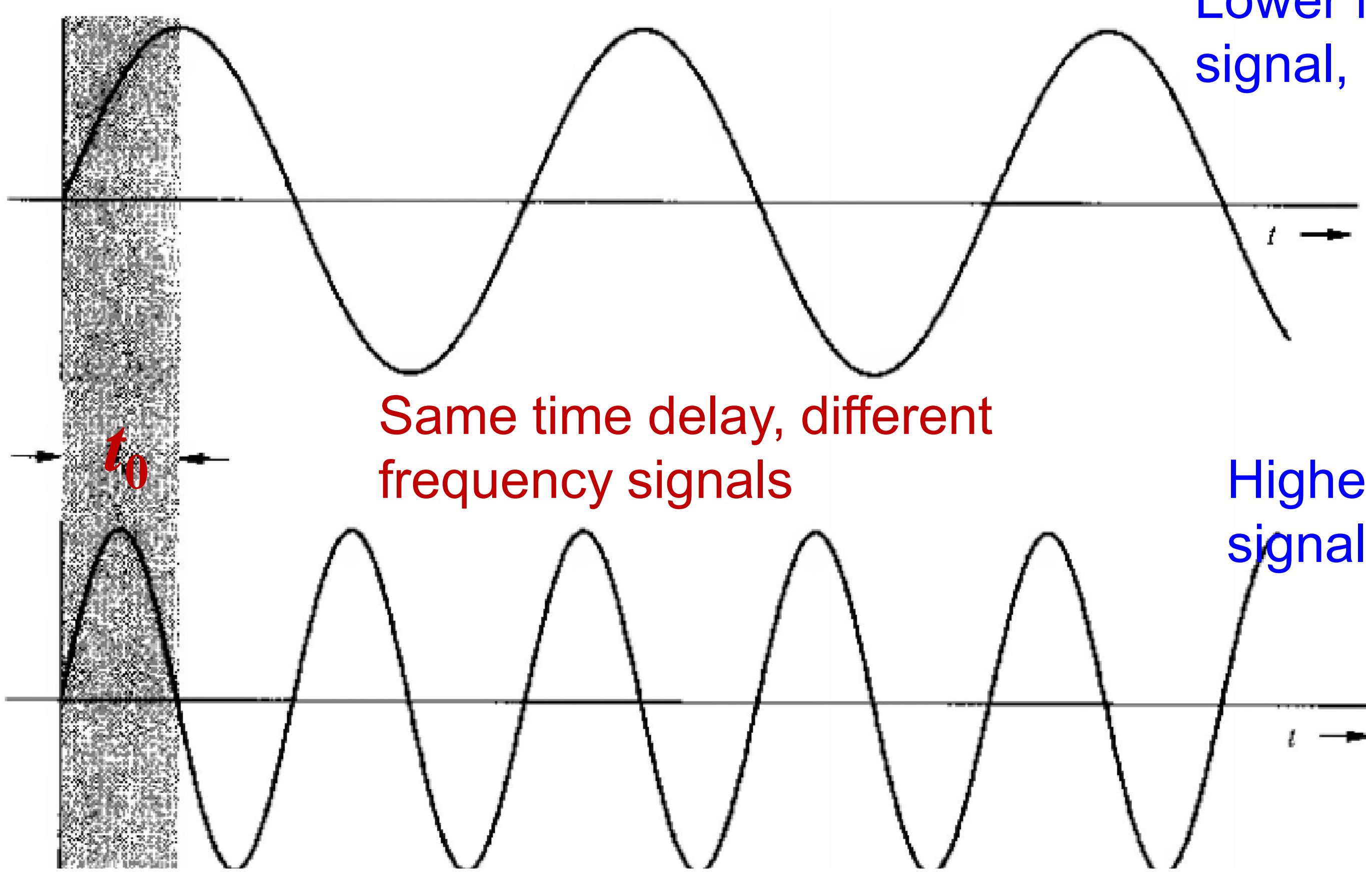


Same time delay, different frequency signals

Higher frequency signal

Properties of Fourier Transform

Time shifting property $g(t-t_0) \Leftrightarrow G(f)\exp(-j2\pi f t_0)$



Lower frequency signal, phase shift $=\pi/2$

Same time delay, different frequency signals

Higher frequency signal, phase shift $= \pi$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Properties of Fourier Transform

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$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$F\left[g(t)e^{j2\pi f_0 t}\right] = \int_{-\infty}^{\infty} g(t)e^{j2\pi f_0 t}e^{-j2\pi f t}dt$$

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$$= \int_{-\infty}^{\infty} g(t)e^{-j2\pi(f-f_0)t}\,dt$$

Properties of Fourier Transform

Frequency shifting property

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$$F\left[g(t)e^{j2\pi f_0 t}\right] = \int_{-\infty}^{\infty} g(t)e^{j2\pi f_0 t}e^{-j2\pi f t}\,dt$$

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$$= G(f-f_0)$$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Replacing f_0 by $-f_0$

$$g(t)\exp(-j2\pi f_0 t) \Leftrightarrow G(f + f_0)$$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Significance:

Multiplying $g(t)$ by $\exp(j2\pi f_0 t)$ shifts the spectrum by f_0 to a new location

Properties of Fourier Transform

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Recall FDM!

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Significance:

Multiplying $g(t)$ by $\exp(j2\pi f_0 t)$ shifts the spectrum by f_0 to a new location

Significance:

As $\exp(j2\pi f_0 t)$ is complex , $g(t)$ is multiplied by a real sinusoid $\cos(2\pi f_0 t)$ where,

$$\cos(2\pi f_0 t) = \frac{1}{2}(e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Therefore, $g(t)\cos(2\pi f_0 t) = \frac{1}{2}\left(g(t)e^{j2\pi f_0 t} + g(t)e^{-j2\pi f_0 t}\right)$

Properties of Fourier Transform

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Therefore, $g(t)\cos(2\pi f_0 t) = \frac{1}{2}(g(t)e^{j2\pi f_0 t} + g(t)e^{-j2\pi f_0 t})$

This results, $g(t)\cos(2\pi f_0 t) \Leftrightarrow= \frac{1}{2}[G(f - f_0) + G(f + f_0)]$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t)\cos(2\pi f_0 t) \Leftrightarrow = \frac{1}{2}[G(f - f_0) + G(f + f_0)]$$

Significance:

- Multiplying a sinusoid $\cos(2\pi f_0 t)$ by $g(t)$ means modulation of sinusoidal amplitude
- $\cos(2\pi f_0 t)$ is the carrier
- $g(t)$ is modulating signal
- This is amplitude modulation

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f-f_0)$$

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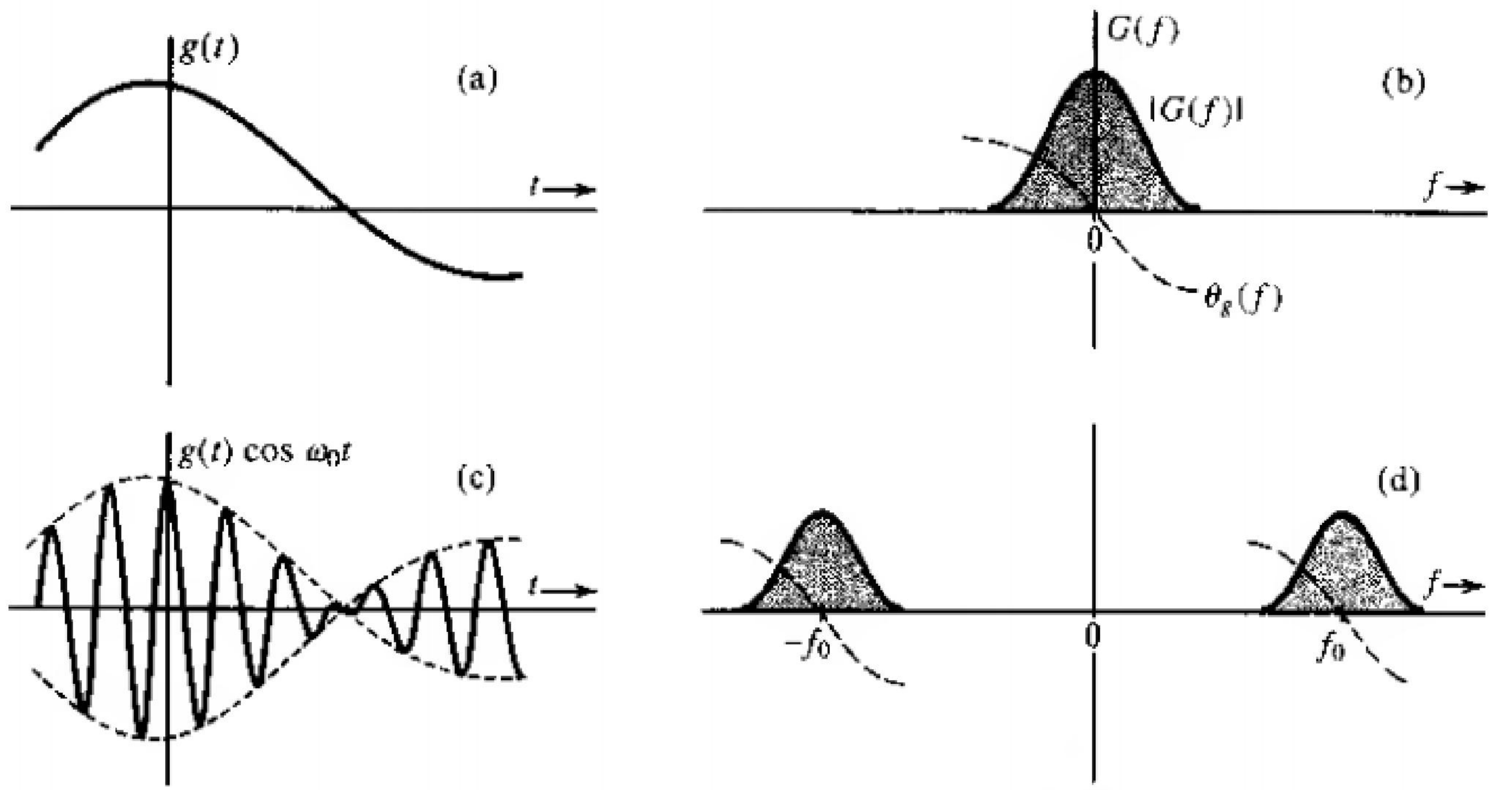

Properties of Fourier Transform

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Further phase change of modulated signal by θ_0:

- Easily done by multiplying $\cos(2\pi f_0 t + \theta_0)$ instead of $\cos(2\pi f_0 t)$ by $g(t)$

Properties of Fourier Transform

Frequency shifting property $\qquad g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$

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$$g(t)\cos(2\pi f_0 t + \theta_0) = \frac{1}{2}[g(t)e^{j2\pi f_0 t + j\theta_0} + g(t)e^{-j2\pi f_0 t - j\theta_0}]$$

$$= \frac{1}{2}[g(t)e^{j2\pi f_0 t}e^{j\theta_0} + g(t)e^{-j2\pi f_0 t}e^{-j\theta_0}]$$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f-f_0)$$

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This means,

$$g(t)\cos(2\pi f_0 t+\theta_0) \Leftrightarrow= \frac{1}{2}[G(f-f_0)e^{j\theta_0} + G(f+f_0)e^{-j\theta_0}]$$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t)\cos(2\pi f_0 t + \theta_0) \Leftrightarrow= \frac{1}{2}[G(f - f_0)e^{j\theta_0} + G(f + f_0)e^{-j\theta_0}]$$

$\theta_0 = 0$ means

$$g(t)\cos(2\pi f_0 t) \Leftrightarrow= \frac{1}{2}[G(f - f_0) + G(f + f_0)]$$

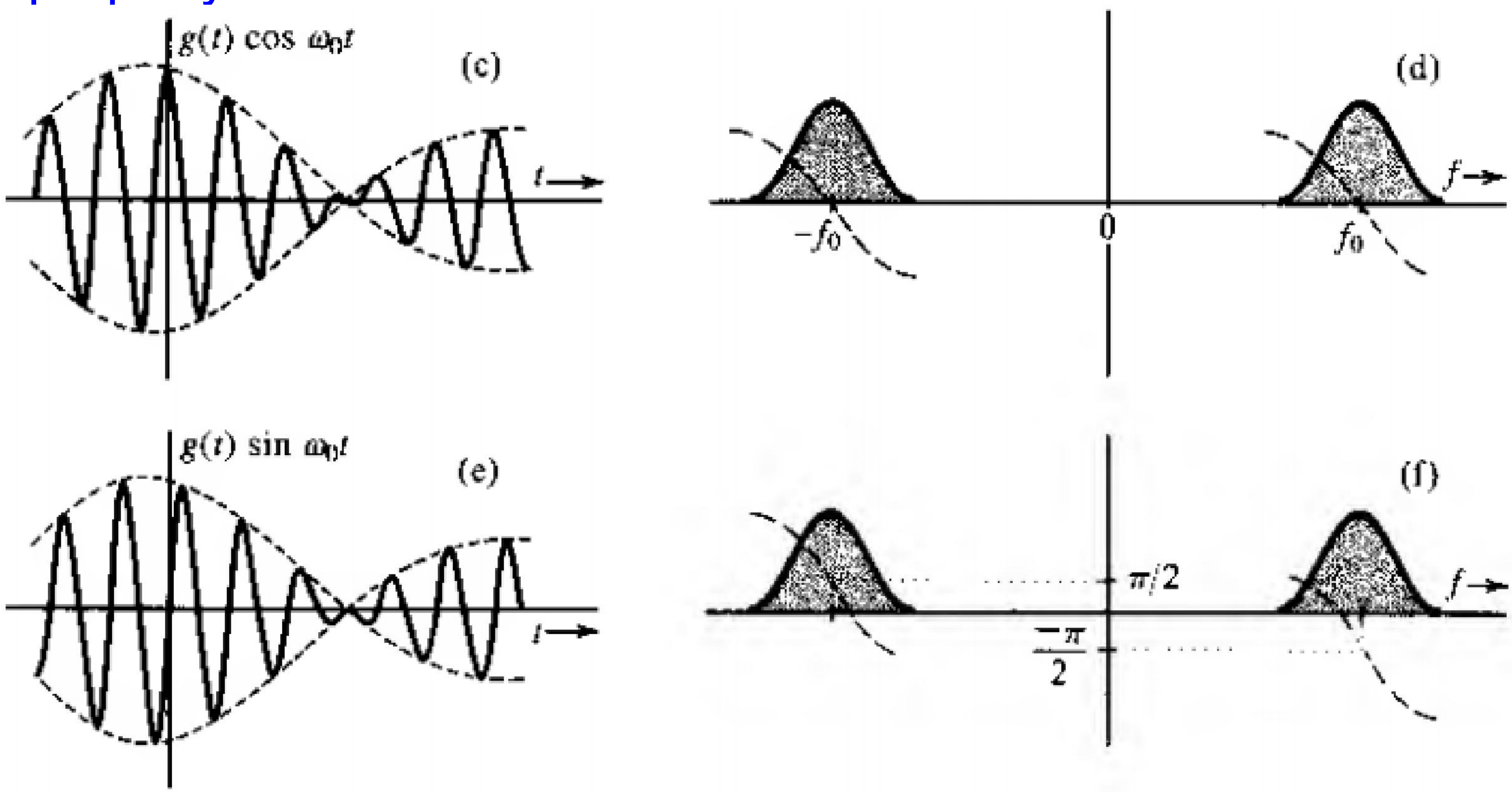
$\theta_0 = -\pi/2$ means

$$g(t)\cos(2\pi f_0 t - \frac{\pi}{2}) \Leftrightarrow= \frac{1}{2}[G(f - f_0)e^{-j\frac{\pi}{2}} + G(f + f_0)e^{j\frac{\pi}{2}}]$$

Properties of Fourier Transform

Frequency shifting property

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f-f_0)$$



Properties of Fourier Transform

Frequency shifting property in Modulation

$$g(t)\exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t)\cos(j2\pi f_0 t) \Leftrightarrow = \frac{1}{2}[G(f - f_0) + G(f + f_0)]$$

Significance:

- A single common channel is used for multiple low band signals (e.g., multiple radio stations)
- Carriers of different frequencies are assigned to different transmission stations
- Each station shifts its signal to its own allocated band
- Simultaneous transmission is NO longer a problem

Properties of Fourier Transform

Convolution and
Modulation Property

Time Convolution

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

Modulation or frequency Convolution

$$g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$$

Properties of Fourier Transform

What is Convolution?

$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t-\tau) d\tau$$

Properties of Fourier Transform

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Properties of Fourier Transform

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$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t-\tau) d\tau = \int_{-\infty}^{\infty} g_2(\tau) g_1(t-\tau) d\tau$$

Properties of Fourier Transform

Time Convolution Property

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

$$\Im\{g_1(t) * g_2(t)\} = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} g_1(\tau)g_2(t-\tau)d\tau\right]e^{-j2\pi ft}dt$$

$$= \int_{-\infty}^{\infty} g_1(\tau)\left[\int_{-\infty}^{\infty} g_2(t-\tau)e^{-j2\pi ft}dt\right]d\tau$$

Properties of Fourier Transform

Time Convolution Property

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$$= \int_{-\infty}^{\infty} g_1(\tau)\left[\int_{-\infty}^{\infty} g_2(t-\tau)e^{-j2\pi ft}dt\right]d\tau$$

$$\int_{-\infty}^{\infty} g_2(t-\tau)e^{-j2\pi ft}dt \text{ is FT of } \tau\text{ - time shifted } g_2(t), e.g., G_2(f)e^{-j2\pi f\tau}$$

Properties of Fourier Transform

Time Convolution Property

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

$$\Im\{g_1(t) * g_2(t)\} = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} g_1(\tau)g_2(t-\tau)d\tau\right]e^{-j2\pi ft}dt$$

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$$= \int_{-\infty}^{\infty} g_1(\tau)G_2(f)e^{-j2\pi f\tau}d\tau = G_2(f)\int_{-\infty}^{\infty} g_1(\tau)e^{-j2\pi f\tau}d\tau$$

$$= G_2(f)\int_{-\infty}^{\infty} g_1(t)e^{-j2\pi ft}dt = G_2(f)G_1(f)$$

Properties of Fourier Transform

Modulation or frequency Convolution Property $g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$

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Modulation or frequency Convolution Property $g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$

$$\Im^{-1}\left\{G_1(f) * G_2(f)\right\} = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} G_1(\tau)G_2(f-\tau)d\tau\right]e^{j2\pi ft}df$$

Properties of Fourier Transform

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Properties of Fourier Transform

Modulation or frequency
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$$= \int_{-\infty}^{\infty} G_1(\tau)g_2(t)e^{j2\pi t\tau}d\tau = g_2(t)\int_{-\infty}^{\infty} G_1(\tau)e^{j2\pi t\tau}d\tau$$

Properties of Fourier Transform

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$$= g_2(t)\int_{-\infty}^{\infty} G_1(f)e^{j2\pi ft}df = g_2(t)g_1(t)$$

Properties of Fourier Transform

Time Integration Property

$$\int_{-\infty}^{t} g(\tau)d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$$

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Recall

$$u(t) = \begin{cases} 1 & \text{if } \; t \geq 0 \\ 0 & \text{if } \; t < 0 \end{cases}$$

Properties of Fourier Transform

Time Integration Property

$$\int_{-\infty}^{t} g(\tau)d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$$

Recall

$$u(t) = \begin{cases} 1 & \text{if } \; t \geq 0 \\ 0 & \text{if } \; t < 0 \end{cases}$$

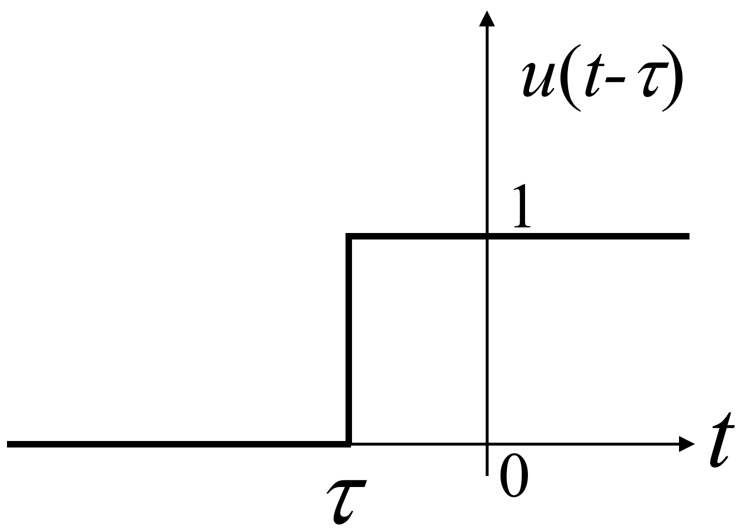
Therefore,

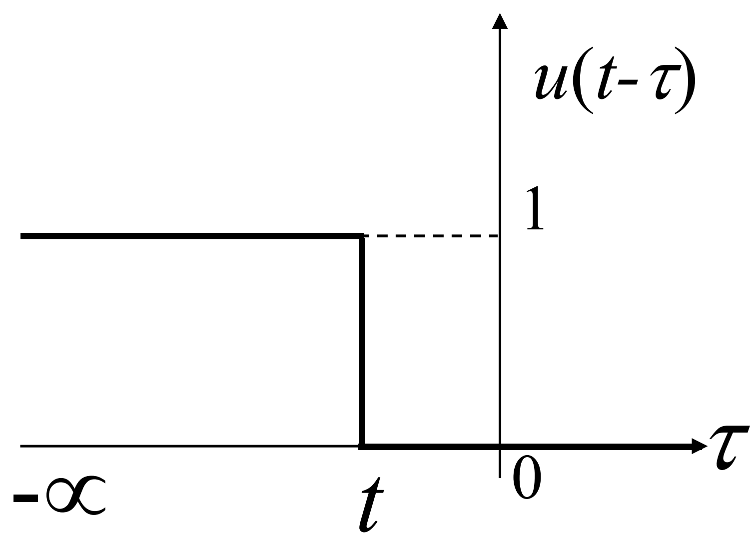
$$u(t-\tau) = \begin{cases} 1 & \text{if } \; t-\tau \geq 0 & \text{or } \; \tau \leq t \\ 0 & \text{if } \; t-\tau < 0 & \text{or } \; \tau > t \end{cases}$$

Properties of Fourier Transform

Time Integration Property

$$\int_{-\infty}^{t} g(\tau)d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$$





$$u(t-\tau) = \begin{cases} 1 & \text{if } \tau \leq t \\ 0 & \text{if } \tau > t \end{cases}$$

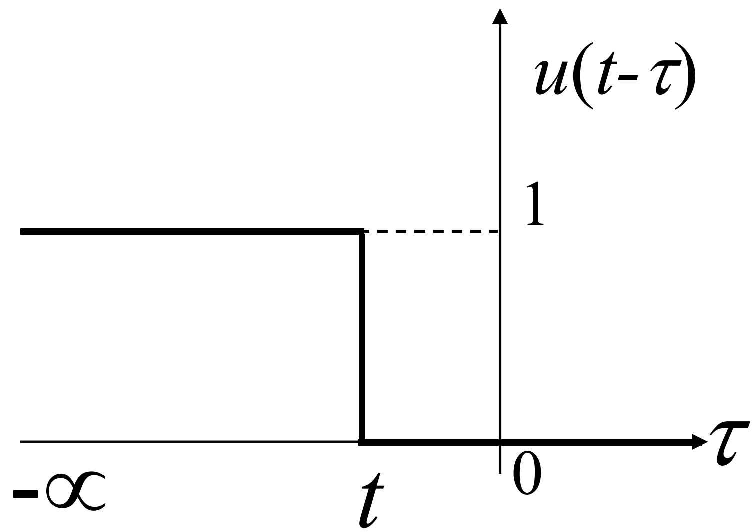
$$u(t-\tau) = \begin{cases} 1 & \text{if } t-\tau \geq 0 \\ 0 & \text{if } t-\tau < 0 \end{cases}$$

Properties of Fourier Transform

Time Integration Property

$$\int_{-\infty}^{t} g(\tau)d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^{\infty} g(\tau)u(t-\tau)d\tau$$



$$u(t-\tau) = \begin{cases} 1 & \text{if } \ t-\tau \geq 0 \\ 0 & \text{if } \ t-\tau < 0 \end{cases}$$

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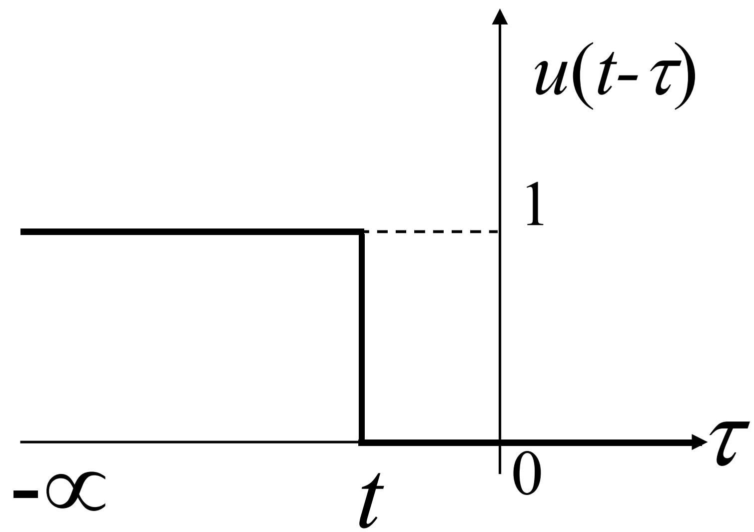

$$u(t-\tau) = \begin{cases} 1 & \text{if } \; t-\tau \geq 0 \\ 0 & \text{if } \; t-\tau < 0 \end{cases}$$

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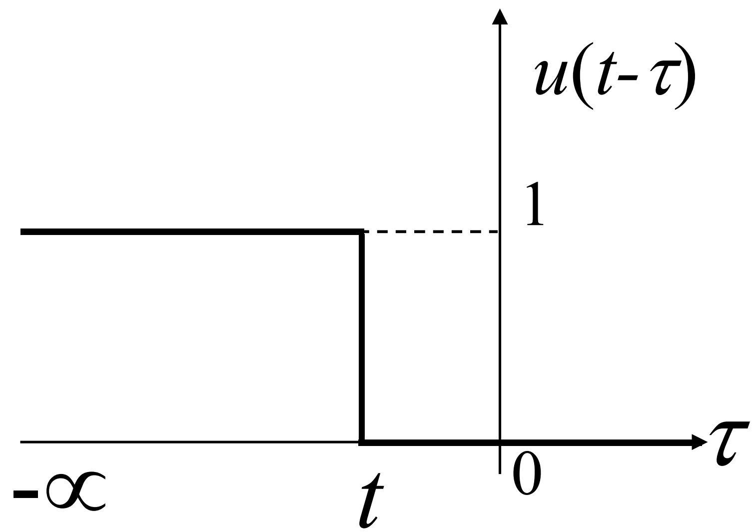
$$= \int_{-\infty}^{t} g(\tau).1d\tau + \int_{t}^{\infty} g(\tau).0d\tau = \int_{-\infty}^{t} g(\tau)d\tau$$



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Properties of Fourier Transform

Time Integration Property

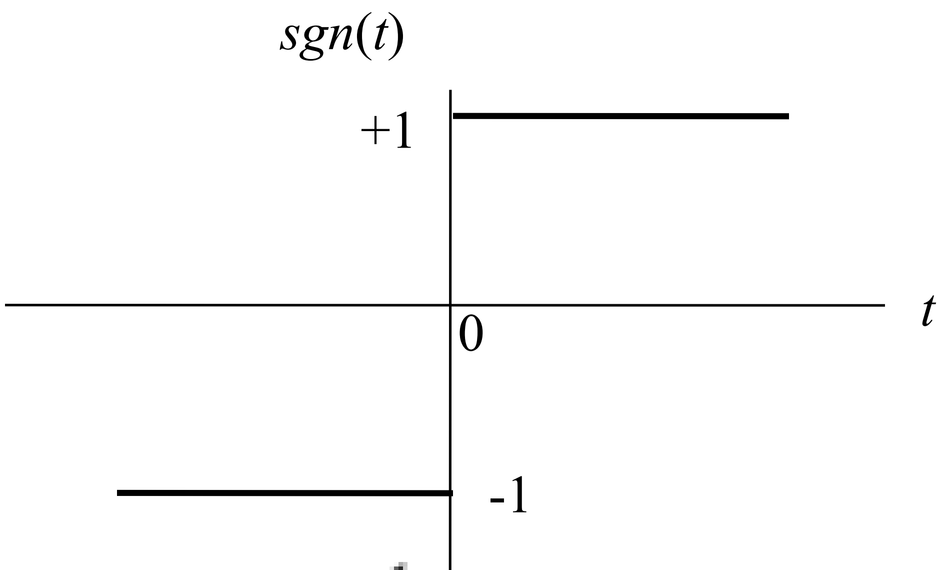
$$\int_{-\infty}^{t} g(\tau)\,d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$$

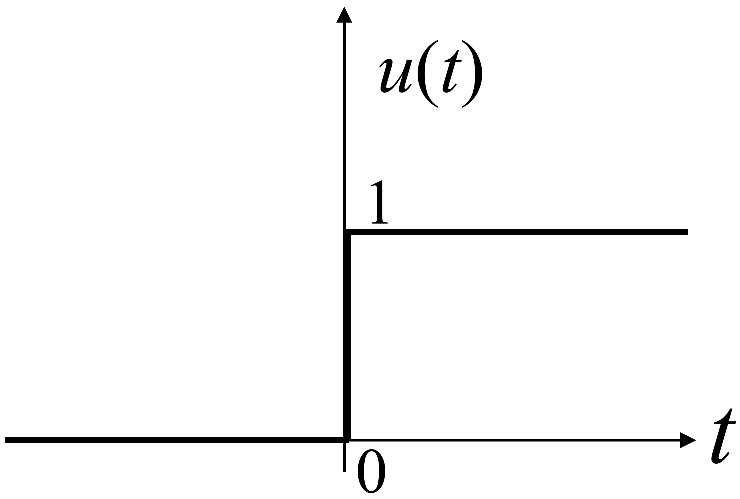
$$g(t) * u(t) = \int_{-\infty}^{t} g(\tau)\,d\tau$$

$$\int_{-\infty}^{t} g(\tau)\,d\tau = g(t) * u(t) \Leftrightarrow G(f)U(f)$$

Properties of Fourier Transform

u(*t*) and *sgn*(*t*) function



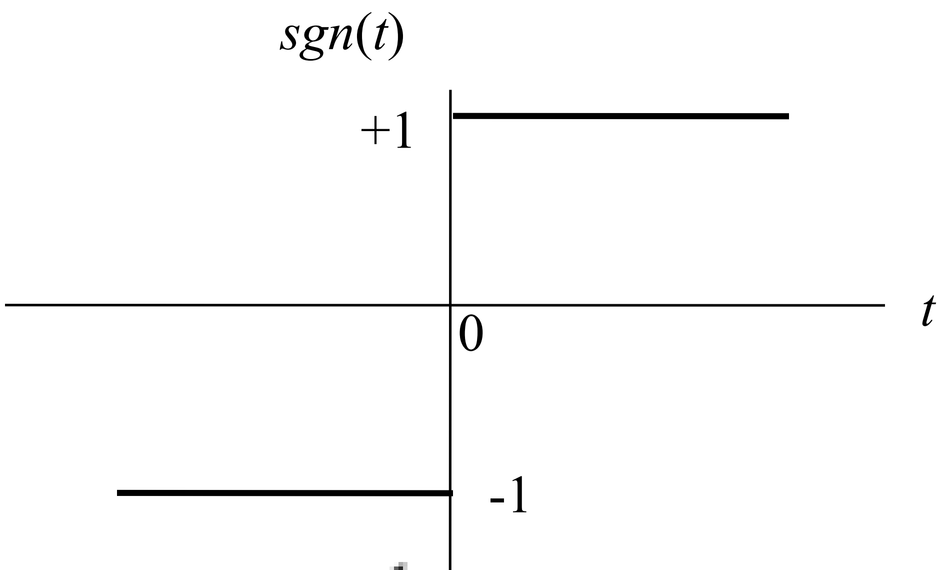


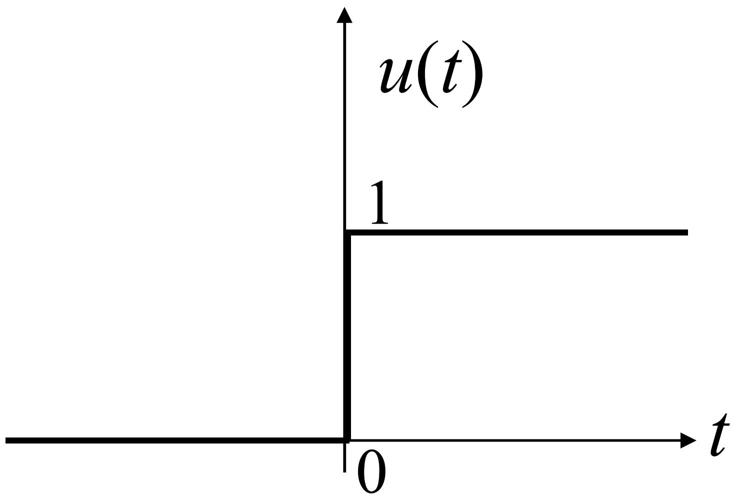
$$\operatorname{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

$$u(t) = \begin{cases} 1 & \text{if } \mathrm{t} \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Properties of Fourier Transform

u(*t*) and *sgn*(*t*) function





$$\mathrm{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Therefore, $u(t) = \frac{1}{2}\left(\mathrm{sgn}(t) + 1\right)$

Properties of Fourier Transform

u(*t*) and *sgn*(*t*) function

$$u(t) = \frac{1}{2}\left(\mathrm{sgn}(t) + 1\right)$$

Therefore, $U(f) = \frac{1}{2}\left(\frac{1}{j\pi f} + \delta(f)\right) = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$

Properties of Fourier Transform

Time Integration Property

$$\int_{-\infty}^{t} g(\tau)d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2}G(0)\delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^{t} g(\tau)d\tau$$

$$\int_{-\infty}^{t} g(\tau)d\tau = g(t) * u(t) \Leftrightarrow G(f)U(f)$$

$$U(f) = \frac{1}{2}\left(\frac{1}{j\pi f} + \delta(f)\right) = \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$$

Properties of Fourier Transform

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$$\int_{-\infty}^{t} g(\tau)d\tau = g(t)*u(t) \Leftrightarrow G(f)U(f)$$

$$=\frac{G(f)}{j2\pi f}+\frac{1}{2}G(f)\delta(f)$$

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Properties of Fourier Transform

Time Differentiation Property

$$\frac{dg(t)}{dt} \Leftrightarrow j2\pi f G(f)$$

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Properties of Fourier Transform

Time Differentiation Property

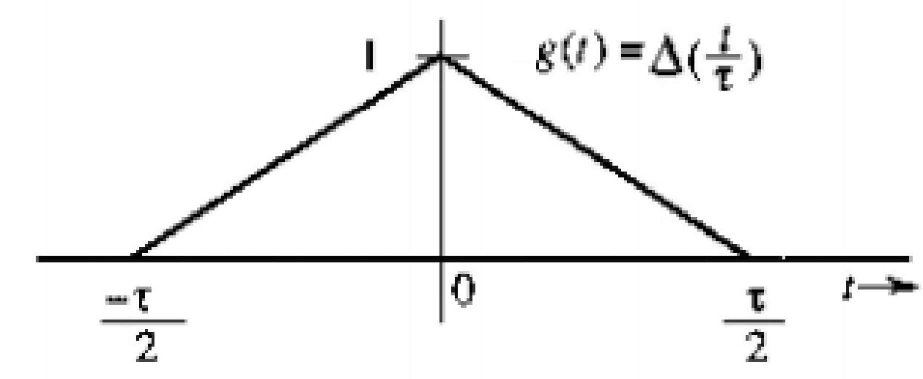
$$\frac{dg(t)}{dt} \Leftrightarrow j2\pi f G(f)$$

$$\frac{d^n g(t)}{dt^n} \Leftrightarrow \left(j2\pi f\right)^n G(f)$$

Properties of Fourier Transform

Time Differentiation Property to find FT of complex signal

$$\frac{d^n g(t)}{dt^n} \Leftrightarrow \left(j2\pi f\right)^n G(f)$$



Properties of Fourier Transform

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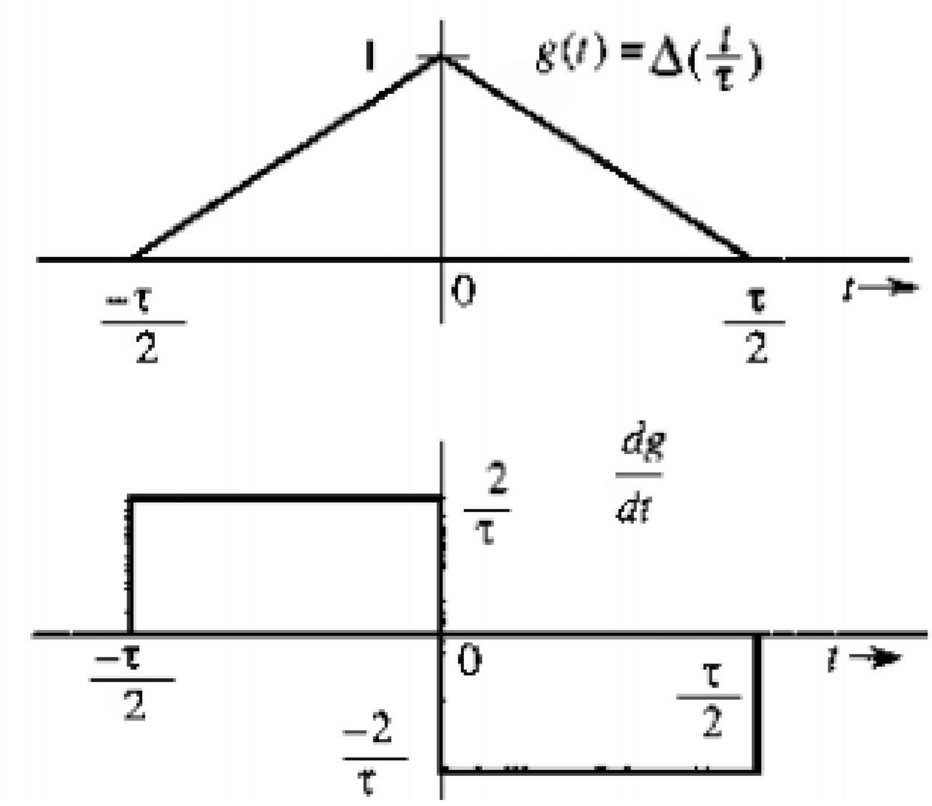

Properties of Fourier Transform

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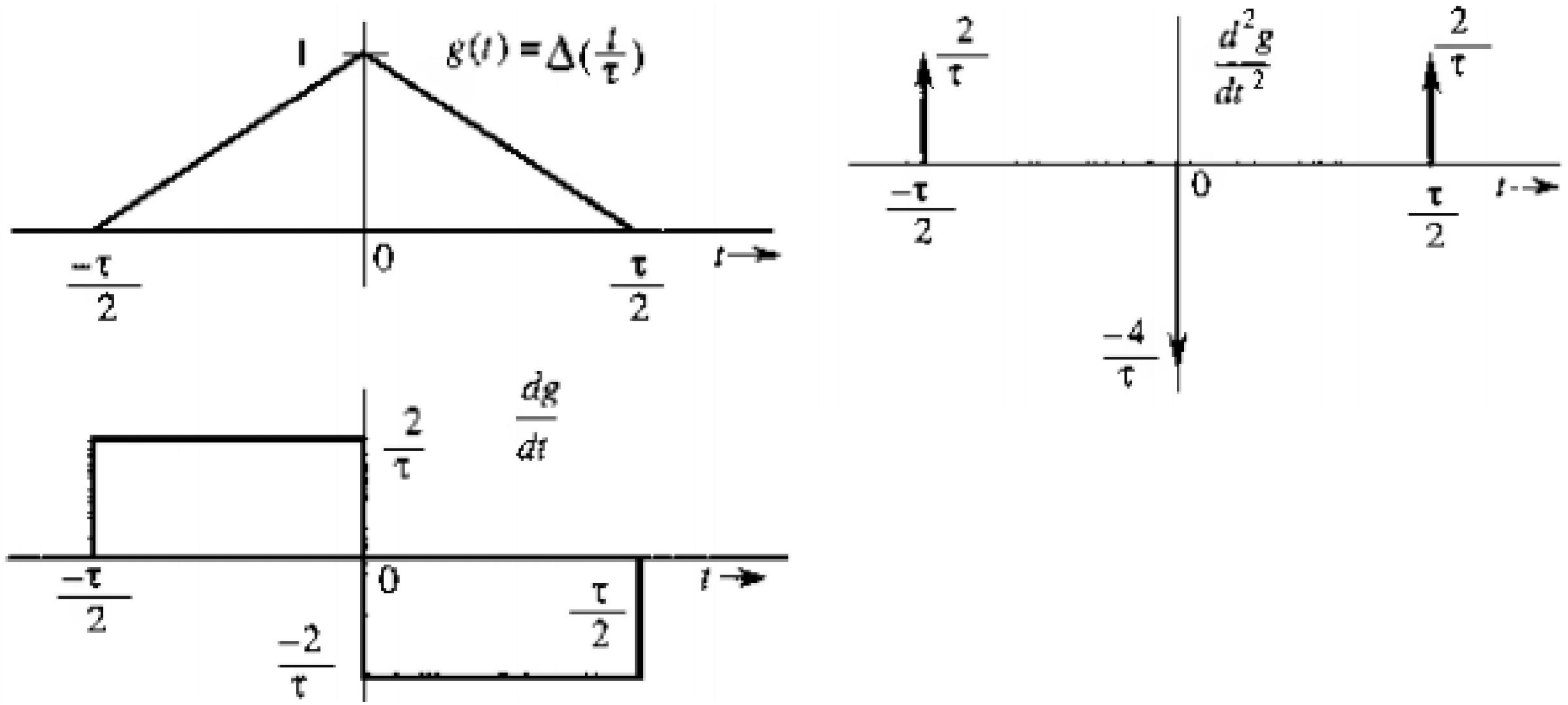

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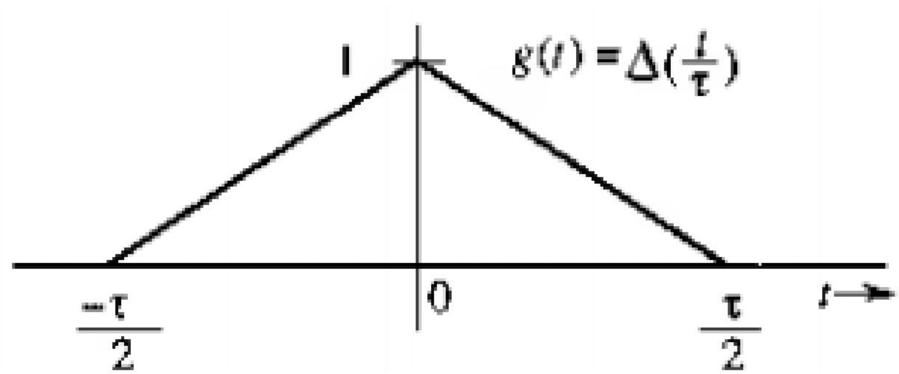
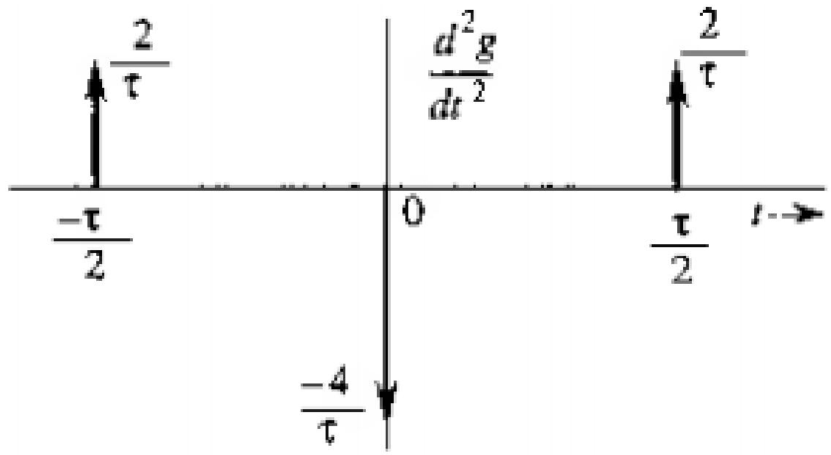


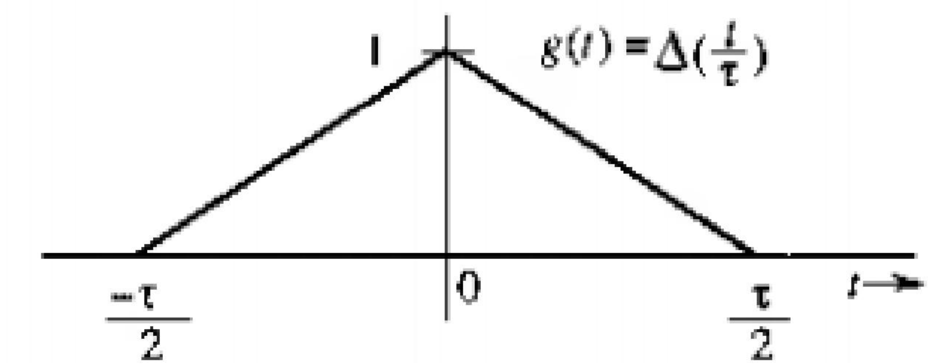


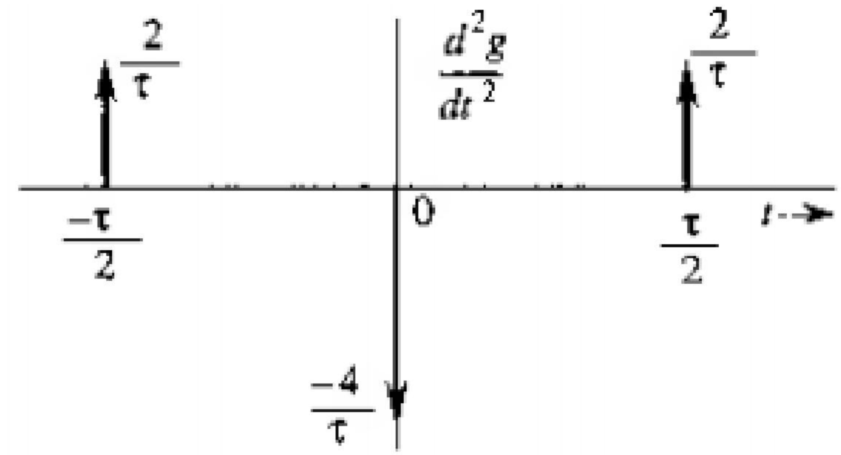
$$\text{Let, } \Im(\mathrm{g(t)}) = G(f)$$

Properties of Fourier Transform

Time Differentiation Property to find FT of complex signal

$$\frac{d^n g(t)}{dt^n} \Leftrightarrow \left(j2\pi f\right)^n G(f)$$





$$\frac{d^2 g(t)}{dt^2} = \frac{2}{\tau}\left[\delta\left(t+\frac{\tau}{2}\right) - 2\delta(t) + \delta\left(t-\frac{\tau}{2}\right)\right]$$

$$\frac{d^2 g}{dt^2} \Longleftrightarrow (j2\pi f)^2 G(f) = -(2\pi f)^2 G(f)$$

Properties of Fourier Transform

Time Differentiation Property to find FT of complex signal

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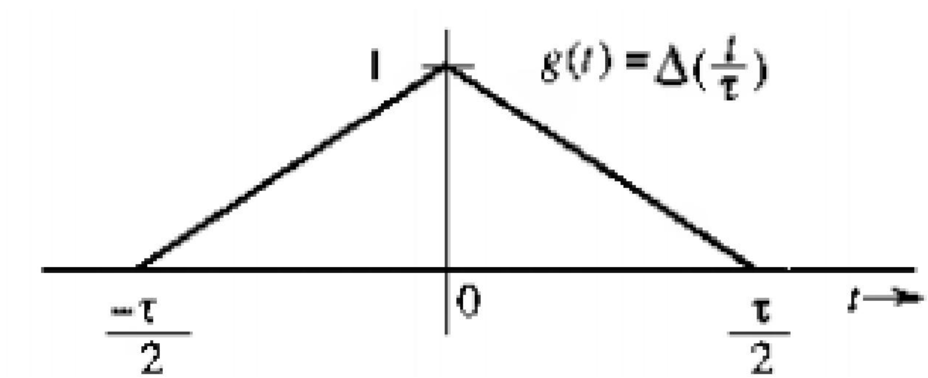

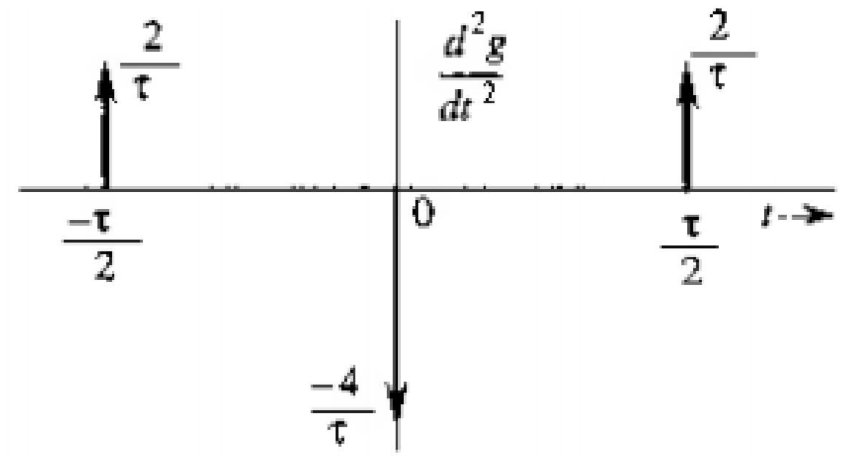

$$\frac{d^2 g(t)}{dt^2} = \frac{2}{\tau}\left[\delta\left(t+\frac{\tau}{2}\right) - 2\delta(t) + \delta\left(t-\frac{\tau}{2}\right)\right] \qquad \delta(t-t_0) \Longleftrightarrow e^{-j2\pi f t_0}$$

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$$\frac{d^2 g}{dt^2} \Longleftrightarrow (j2\pi f)^2 G(f) = -(2\pi f)^2 G(f)$$

$$(j2\pi f)^2 G(f) = \frac{2}{\tau}\left(e^{j\pi f \tau} - 2 + e^{-j\pi f \tau}\right)$$

$$G(f) = \frac{\tau}{2}\operatorname{sinc}^2\left(\frac{\pi f \tau}{2}\right)$$

Properties of Fourier Transform

Time Differentiation Property to find FT of complex signal

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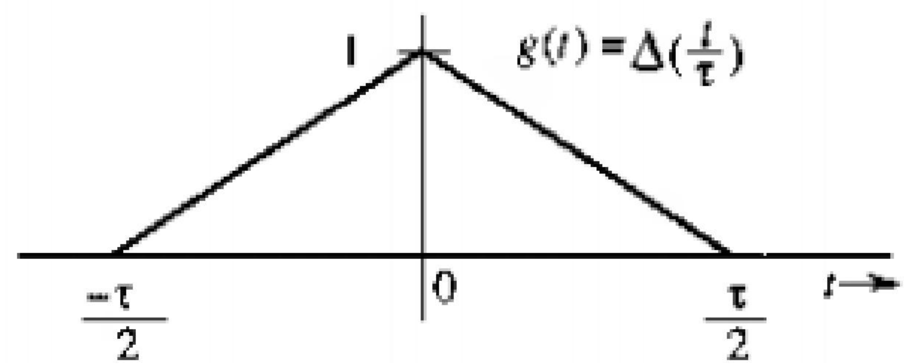
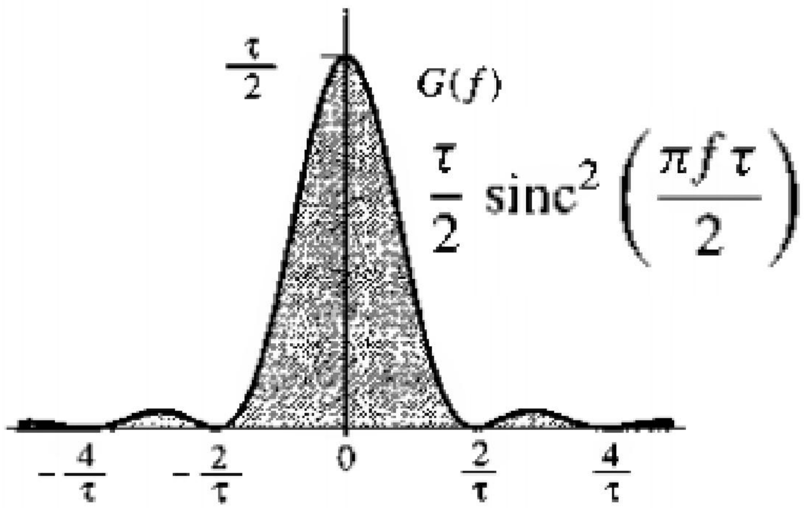
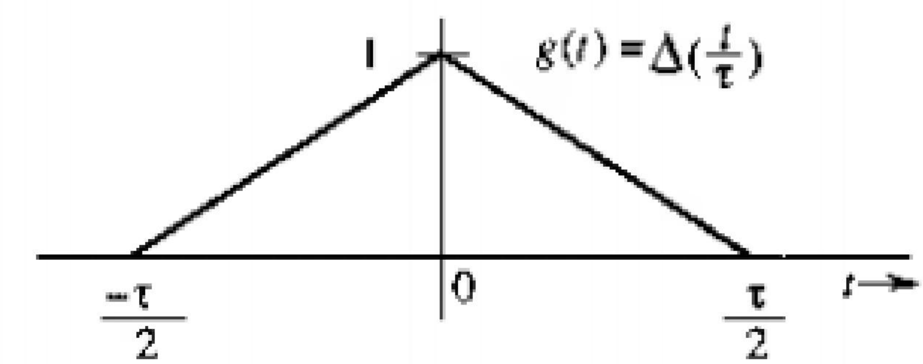
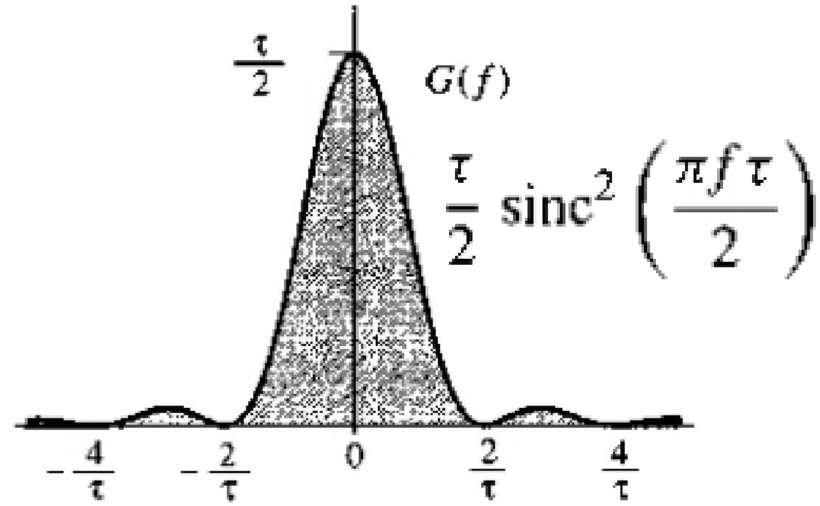




Time Domain and Frequency Domain Representation

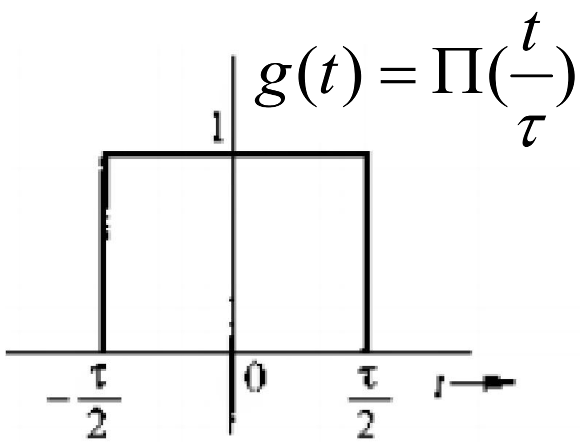


Limited in time

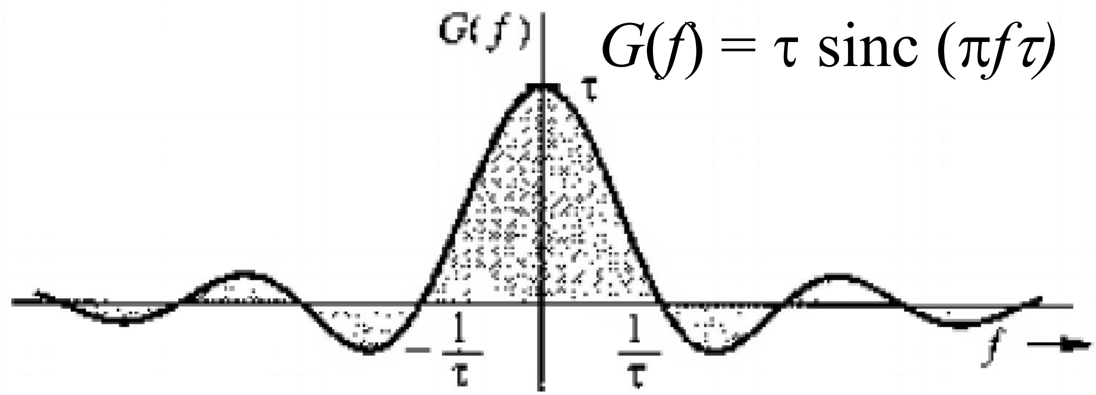


Indefinite spread in frequency

Time Domain and Frequency Domain Representation



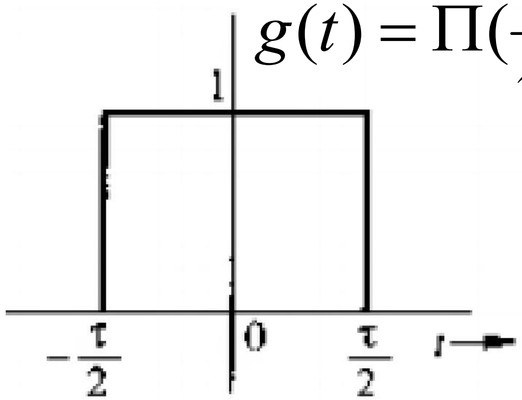
Limited in time



Indefinite spread in frequency

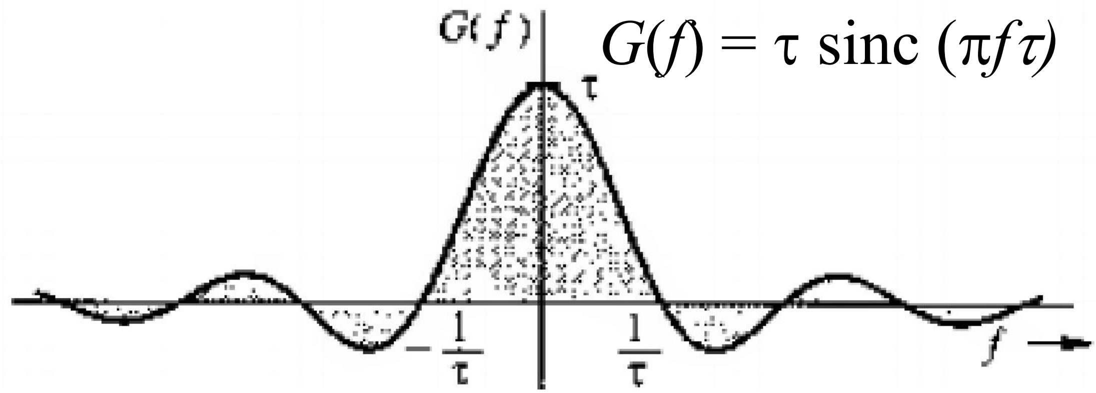
Time Domain and Frequency Domain Representation

$$g(t) = \Pi(\frac{t}{\tau})$$



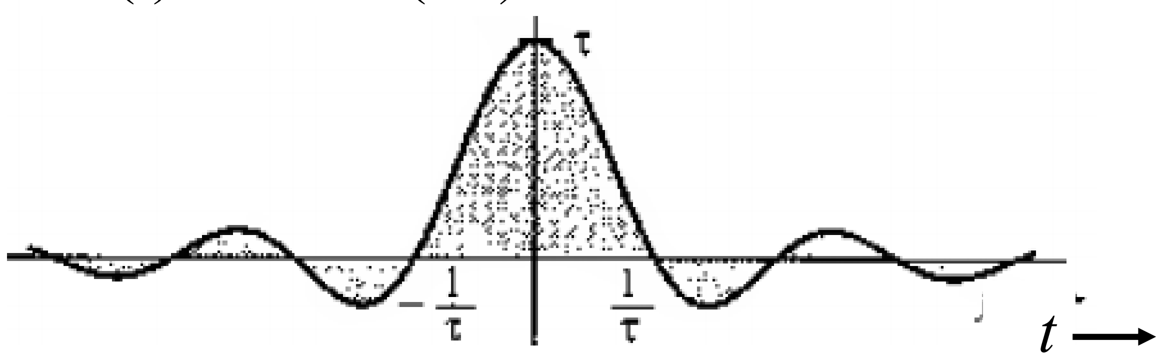
Limited in time

$$G(f) = \tau \text{ sinc } (\pi f \tau)$$



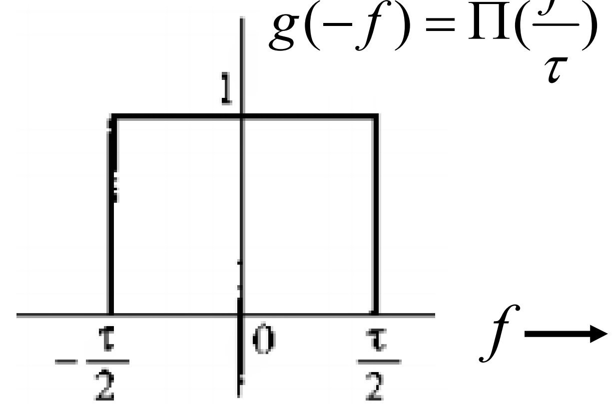
Indefinite spread in frequency

$$G(t) = \tau \text{ sinc } (\pi \tau t)$$



Indefinite spread in time

$$g(-f) = \Pi(\frac{f}{\tau})$$



Limited in frequency

Time Domain and Frequency Domain Representation

- A signal cannot be BOTH band limited and time limited
- Relation is inverse
- Relation changes in inverse manner
- Any change in time specification changes spectrum speciation inversely
- We can specify EITHER of them, NOT BOTH of them